

Explainable Disentangled GAN for Controllable Brain Tumor MRI Synthesis and Counterfactual Analysis

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Abstract — Deep Learning has shown promising results in medical imaging, especially for identifying brain tumors in MRI scans. However, many of these methods operate as "black boxes," making it complex to really understand how they decide their conclusions, which can undermine trust in their results. To address this, our paper introduces an Explainable Generative Adversarial Network (XGAN) designed to create and analyze brain tumor MRI scans. This novel architecture combines a CNN classifier with a generative adversarial network. It learns to separate the underlying features of brain tumor MRIs, modeling the brain's anatomy and the tumor's characteristics independently. This allows us to generate synthetic brain tumor MRIs that maintain consistent anatomy while allowing for controlled variations in tumor pathology. Furthermore, we incorporate explainability techniques like Grad-CAM and counterfactual difference visualization to highlight the specific regions of the brain associated with the tumor. Our experiments, conducted on a brain MRI dataset, demonstrate a high classification accuracy of 96%, with excellent precision and recall for both tumor and non-tumor classifications.

Keywords — *Medical Imaging, Brain Tumor Detection, Explainable AI (XAI), Generative Adversarial Network (GAN), Explainable GAN (XGAN), Convolutional Neural Network (CNN), Feature Disentanglement, Synthetic Data Generation, Grad-CAM, Counterfactual Explanation, Interpretability*

I. INTRODUCTION

Among some of the most difficult diseases to treat is a brain tumor, and for an accurate diagnosis, it is crucial to identify and diagnose such illnesses. For the purpose of diagnosis, MRI is often used for detecting brain tumors. With the help of MRI scans, the structure and abnormalities of the human brain can be visualized in sufficient detail. Manually analyzing MRI scans is not only time-consuming but also tedious and may result in errors, especially while handling a large number of medical images. Hence, it becomes important for an automated system to diagnose brain tumors.

With techniques from Deep Learning like convolutional neural networks (CNNs) [19], they have depicted significant improvements in medical image classification tasks in recent times. The main benefit of using deep learning algorithms lies in the fact that they can learn features from raw MRI scans and

classify them accurately. Simultaneously, the use of deep learning techniques in diagnosing medical conditions was frowned upon due to their so called "black-box" nature in which it was difficult to understand the reasoning behind the output. This poses a challenge to developing an automated system for medical image classification as the opaque nature of the system will hinder its adoption in practical scenarios.

To solve this, Explainable Artificial Intelligence (XAI) has become popular. Tools such as Grad-CAM have proved to be useful in highlighting regions of interest within an input image and helping clinicians determine whether the model uses the correct part of the structure. [10]. Although XAI is becoming increasingly popular, most techniques proposed in literature are only capable of providing post-hoc analysis and fail to provide information regarding the internal representations of the features of the disease.

Within this work, a novel framework is put forward that is based on the idea of using a CNN-based tumor classifier in combination with a Generative Adversarial Network (GAN) to develop an explainable generative model, which is capable of learning a disentangled latent representation in which the anatomical structure and tumor pathology are encoded separately, allowing the tumor characteristics to be manipulated in the generated MRI images and visualized. The proposed system is capable of providing accurate tumor detection and visualization of the tumor features.

Despite the success of GANs in the application of image synthesis and augmentation in medical image processing, it has been noted that the majority of existing GAN-based solutions lack the ability to manipulate disease-related features in an explicit manner [14] [15]. Existing GAN solutions use a single latent space to encode anatomical information and pathological information, which makes it hard to manipulate tumor-related features independently [7] [8]. This makes the generated images look realistic but lack transparency. Moreover, interpretability techniques are only applicable to classification models, while generative models are still black boxes. This paper proposes a disentangled generative model that separates anatomical information from pathological information in the latent space, thus enabling the manipulation of tumor-related features while

keeping anatomical information constant. This enables counterfactual visualization of tumor progression.

The key contribution of the research can be outlined as follows. Firstly, the disentangled generative model architecture is introduced, which decouples the anatomical structure and tumor pathology in the shared latent space. This allows for the interpretable manipulation of the disease features. Secondly, the classifier-guided training mechanism is introduced, which imposes semantic consistency between the generated MRI images and tumor labels. Thirdly, the framework allows for the controlled manipulation of the tumor intensity using the pathology latent vector. Finally, the use of the Grad-CAM explanations and counterfactual difference visualization allows for the meaningful insight of the tumor feature influence.

II. LITERATURE SURVEY

Several studies were carried out to investigate how deep learning can be used in identifying tumor regions of brain from MRI images. The CNN technique was the most popular one adopted in such analysis owing to its effectiveness in automatically learning hierarchical features from the data [19] [25]. Previous works were primarily focused on classification problems, in which CNN-based models were designed to classify brain tumor and non-tumor MRI scans with high accuracy. Though the techniques were successful in achieving the objectives, the techniques were less transparent and could not be applied in real-world scenarios.

Recently, Explainable AI methods have been developed including the technique Grad-CAM which attempts to demystify the techniques and provides an indication of the parts of the image used for making the classification [10][11]. Though the techniques provide useful insights into the brain tumor detection outcome, the techniques are less effective because the techniques could not provide the necessary control over the appearance of the features in the resulting image.

References [1] and [4] demonstrate the application of Generative Adversarial Networks in medical image production and data augmentation. GAN-based models use a generator model to create images, while another model, called the discriminator, classifies the images created by the generator model. As demonstrated in references [13] and [16], GAN-based models have been utilized to produce realistic MRI pictures, improving the training dataset, particularly when coping with a shortage of significant amounts of labeled medical images.

Recent studies have also introduced GAN-based frameworks that are explainable, especially when incorporating interpretability, as seen in reference [12]. Specifically, a generative model has been proposed, where the latent space is composed of features representing anatomical structures and pathology, as seen in reference [21]. In particular, medXGAN [26] proposes a generative framework in which the latent space is explicitly divided into components representing anatomical structure and disease pathology [21]. Through the disentanglement of the factors, the model allows for the controlled manipulation of the pathological features while keeping the anatomical structure constant. Such approaches also promote better understanding of the impact of the disease-

related features on the model's prediction. It is also an emerging trend for the generation of medical images.

Problem Statement: Recently, deep learning models have demonstrated exceptional accuracy in medical picture classification tasks, including the categorization of brain cancers. However, most deep learning models are black box models, meaning that they offer no insight into how the decision is being made, which is a major problem when using such models in clinical settings, as reliability is of utmost importance in such settings. Furthermore, conventional explainability techniques do not provide any control over the pathological elements ingrained in the model's decision-making process, but they do highlight the areas of the image that are crucial for the decision. As a result, it is essential to create a model that not only detects cancers with high accuracy but also provides information on how the tumor's characteristics affect the model's judgment. The goal of this research is to create an explainable generative model that provides controlled creation of MRI images with different tumor intensities, i.e., controlled separation of tumor pathophysiology and anatomical structure in the generative model's latent space.

III. METHODOLOGY

A. Proposed Contribution

The main novelty of this research is the fusion of the concepts of disentangled representation learning [7] [8], classifier-guided generative modeling, and explainability techniques in a unified framework for brain tumor image synthesis from MRI scans. Unlike conventional GAN models, in which the latent vector is considered a single entangled representation, the proposed model considers the latent vector to be composed of two separate components representing the anatomy of the brain and the pathological characteristics of the tumor. This enables the model to regulate both the tumor's intensity and the resultant images' anatomical consistency. Additionally, the suggested model incorporates a CNN classifier that is frozen during the training of the GAN model and guarantees that the output images maintain the tumor's semantic properties. The proposed model also includes the concept of counterfactual image generation, in which the pathological characteristics of the tumor in the latent vector are modified to visualize the appearance or disappearance of the tumor while keeping the anatomy of the brain the same. Lastly, to make sure the model is concentrating on the pertinent aspects of the image, the suggested model applies Grad-CAM-based explanation mechanisms to the outputs of the CNN classifier and the GAN model.

B. Explainable Counterfactual Analysis

The use of counterfactual analysis to the explanation of tumor generation behavior is one of the main components of the suggested approach. As mentioned earlier, counterfactual explanations [23] involve the generation of alternative samples, where one attribute is varied while other attributes remain the same. In the context of the proposed framework, counterfactual MRI images are generated, keeping the anatomical part constant, while the pathology part is varied. Such pairs of images have the same brain structure, but the tumor conditions are varied. By computing the difference maps between the two

images, the tumor features introduced due to the pathology vector can be identified. Such counterfactual analysis helps in understanding the behavior of the generative model in terms of tumor representation. Moreover, this helps in ensuring that the tumor features are correctly identified, rather than being randomly distributed over the entire image.

C. Classifier-Guided Semantic Regularization

To guarantee that the generated MRI images have meaningful features related to tumors, the proposed framework has been equipped with a classifier-guided semantic regularization mechanism. Once the CNN tumor classifier has been trained, its weights are incorporated into the generative training process. The CNN classifier classifies synthetic pictures produced by the CNN generator during each generative training iteration. This classification loss is combined with the adversarial loss used for training the generative adversarial network [1].

The adversarial loss is defined as:

$$L_a dv = E \left[\log \left(1 - D(G(z)) \right) \right]$$

The classification loss is computed using binary cross-entropy. It is defined as:

$$L_c ls = BCE(C(G(z)), y)$$

where C represents the pretrained frozen classifier and y is the target tumor label.

Therefore, the total generator loss is defined as:

$$L_G = L_a dv + \lambda_c ls * L_c ls$$

In this work, $\lambda_c ls$ is set to 1.0, providing equal weighting between adversarial realism and classification consistency.

D. Controllable Pathology Manipulation

The capacity to regulate the tumor's intensity within the generated images using the MRI scans is a key component of the suggested architecture. This is because the intensity of the pathological vector will directly influence the intensity of the tumor within the generated images, given the division of the latent representation into anatomical and pathological components [7]. By adjusting the intensity of the pathological vector while keeping the anatomical vector intensity constant, the proposed framework will generate a range of MRI images from normal brain scans to images with intensified tumor structures. This ability to control the pathological information within the generated images will provide a useful tool for demonstrating the interpretability of the pathological information within the generated images.

E. Dataset Overview

To increase variety and lessen dataset bias, several publicly accessible brain MRI datasets were combined to create the dataset used in this study. The primary sources include Brain Tumor MRI Dataset (Masoud Nickparvar, Kaggle), Brain Tumor Dataset Version 8 (Jun Cheng), Brain Tumor Classification Dataset (Sartaj, Kaggle), Brain Tumor MRI Images - 44 Classes (Fernando Feltrin, Kaggle), Brain Tumors 256×256 Dataset (Thomas, Kaggle) and Brain Tumor MRI

Scans (Rajarshi Mandal, Kaggle). All images were converted to grayscale, resized to 128×128 resolution, and normalized to the range [-1, 1]. A consistent preprocessing pipeline was applied across all datasets to ensure uniformity.

The final dataset comprises 4,254 tumor images and 4,187 non-tumor images, which is essentially a balanced binary classification dataset. Combining multiple datasets guarantees greater tumor class diversity, which enhances the model's capacity for generalization.

After preprocessing the images, the dataset is further divided into two categories comprising of training and validation sets using the stratified method.

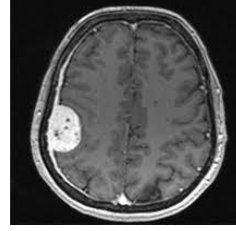


Fig. 1: Tumor MRI

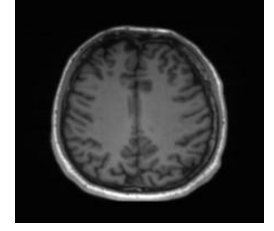


Fig. 2: No-Tumor MRI

F. Dataset Preparation

To enhance the quality and applicability of the data, preprocessing techniques were applied to deep learning models prior to model training. To reduce data dimensionality, MRI images were converted to grayscale format, which is considered appropriate for this type of data. Normalization of pixel values between -1 and 1 was conducted, which is considered appropriate for training models using the GAN technique, thereby stabilizing generator and discriminator networks.

Further, the images were reshaped to accommodate an additional dimension, which is necessary for CNN models. Therefore, the final shape of the data was 128×128×1. Random sampling and visualization were employed to validate the accuracy of the preprocessing techniques and ascertain that both tumor and non-tumor data were included in the dataset.

G. Model Architecture

There are two primary modules in the suggested model: a tumor detector that uses a CNN and image generation through Generative Adversarial Networks (GAN). The classifier is based on a CNN and detects tumors present in the MRI scans. To extract features from the MRI scans, the classifier architecture consists of convolution layers [19], batch normalization, and pooling layers.

After that, the features are fed into the dense layer, which classifies binary classes using the sigmoid activation function, and the global average pooling layer. The GAN architecture, which consists of the discriminator and generator networks, served as the model's inspiration. The generator network is responsible for generating artificial MRI images, whereas the discriminator network evaluates the artificial images' authenticity.

To ensure that the generated synthetic images are meaningful and interpretable, the latent vector is split into two

parts, which represent the anatomical structure of the brain and the tumor pathology, respectively.

$$z = [z_a, z_p]$$

where $z_a \in R^{80}$ represents anatomical features and $z_p \in R^{20}$ represents pathological features. This separation enables controlled manipulation of tumor characteristics while preserving anatomical consistency, where z_a represents anatomical features and z_p represents pathological features.

H. Model Training

The CNN model was trained using the Adam optimizer with a learning rate of 0.0001. The issue of overfitting has been mitigated by making use of the early stopping approach during training. Training for the CNN model was conducted till 80 epochs with 16 as the batch size. For implementation of the CNN model, TensorFlow was used in a NVIDIA GPU environment.

The generator “G” and discriminator “D” models compete against one another in a minimax game of optimization in the adversarial learning approach used to create the GAN model [1].

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim \mathbb{P}_{dt}(x)} [\log D(x)] + \mathbb{E}_{z \sim \mathbb{P}_z(z)} [\log (1 - D(G(z)))]$$

While the discriminator attempts to distinguish between the genuine and fake images, the generator attempts to create images that mimic the MRI scans during the model's training. Apart from the adversarial learning process, there is also the use of classification-guided loss on the generator to ensure semantic consistency of the tumor characteristics.

I. Model Evaluation

Accuracy, precision, recall, and F1 score were among the classification performance metrics used to evaluate this model's performance. Standard classification measures were employed

to assess the suggested model's performance quantitatively. These metrics are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP, TN, FP, and FN stand for true positives, true negatives, false positives, and false negatives, in that order.

Accuracy is used to determine the percentage of correctly classified images, whereas recall along with precision provide deeper insight into how well the model is performing for tumor and non-tumor classification. In addition to classification performance measures, visualization techniques such as Grad-CAM [10] were used to evaluate the model's interpretability performance.

The system's performance is based on a multi-stage approach that integrates classification, generative, and explainable models. Initially, MRI images are preprocessed and used to train a convolutional neural network that is capable of performing classification between tumor and non-tumor brain image classification. Once trained, the model achieves high predictive performance, after which it is frozen to act as a semantic guidance component for the generative component. During training, the generator component is trained on a vector that is divided into anatomical and pathological components. The anatomical component is responsible for capturing features related to structural characteristics, whereas the pathological component is mainly considered for capturing features related to tumor characteristics. The generator component is responsible for generating synthetic MRI images, whereas the discriminator

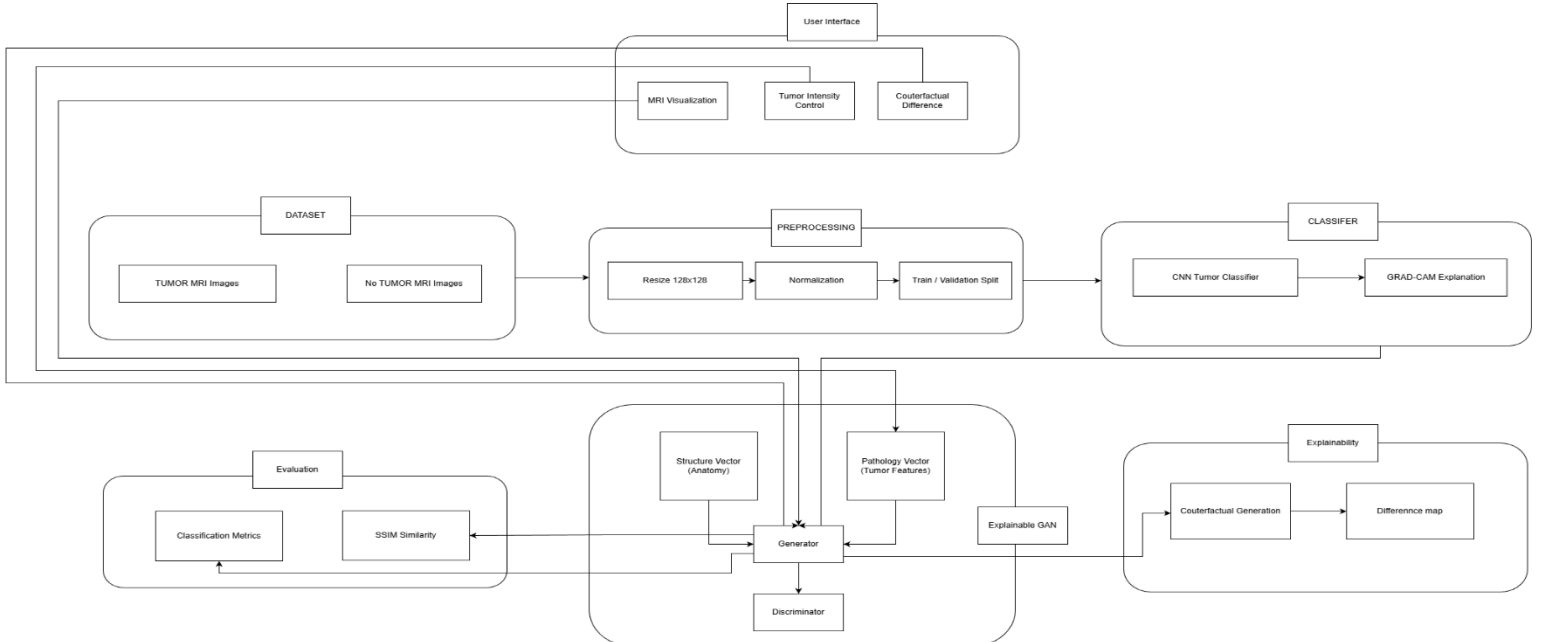


Fig. 3: Architecture Diagram

component is responsible for determining whether the synthetic image is realistic. In addition, the generator component is trained on classification loss, which is derived from a CNN classifier, which is then used to generate synthetic images that correspond to tumor characteristics. This approach forces the generator component to learn features related to tumors, which are then embedded into synthetic images.

The performance is based on explainability mechanisms such as Grad-CAM visualization and counterfactual difference mapping, which provide deeper insight into how anatomical structure, tumor, and model performance relate to each other [23].

The Grad-CAM localization map is computed as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right)$$

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

Where α_k^c is importance weight for feature map k for class c , A_{ij}^k is spatial location activation (i, j) in feature map k , y^c is score for class c and Z is normalization factor (number of pixels in the feature map)

IV. RESULTS

The CNN classifier showed good results in the differentiation of tumor and non-tumor MRI images. The classifier brought result of 96% accuracy overall on the validation dataset, which included 817 MRI images.

The classifier's results, including precision, recall, and F1-score for both classes, are displayed in the table below.

Class	Precision	Recall	F1-score	Support
No Tumor	0.98	0.95	0.96	466
Tumor	0.93	0.98	0.96	351
Accuracy			0.96	817
Macro-Avg	0.96	0.96	0.96	817
Weighted-Avg	0.96	0.96	0.96	817

Table. 1: Evaluation Metrics

The classifier also ensures performance balanced for the two classes, with the F1-score for both tumor and non-tumor types at 0.96, which ensures that the classifier is reliable for detection.

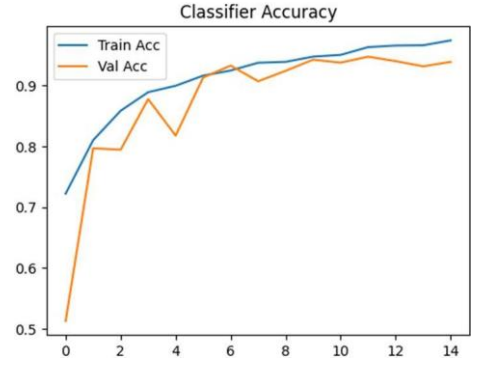


Fig. 4: Classifier Accuracy

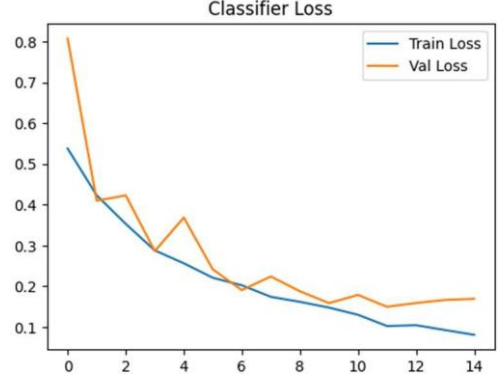


Fig. 5: Classifier Loss

In addition, the confusion matrix provides information about the performance of the algorithm, which can be calculated based on the actual classification and the predicted classification. It can be seen that the classifier is capable of detecting most of the tumors. There are only a few cases of false negatives (7 instances), where the images of tumors are incorrectly classified as not being tumors. The images that resulted in the false negative cases were seen to have tumors that had low contrast, small size, and blurred boundaries.

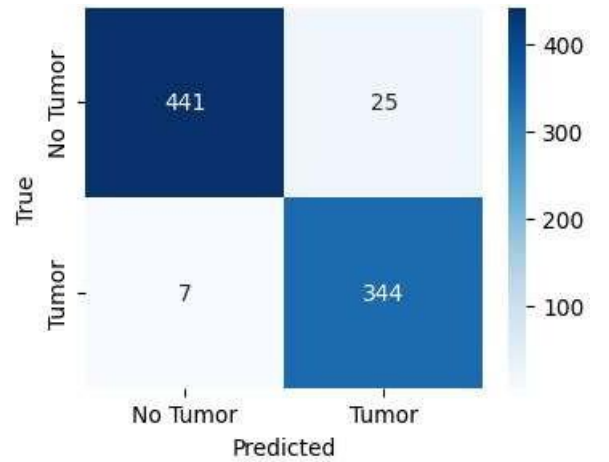


Fig. 6: Confusion matrix

Latent Pathology Manipulation Results

Experiments have been carried out to investigate how modifications to the pathology vector impact the final MRI picture in order to determine the effectiveness of the disentangled representation. By fixing the anatomical vector and scaling up the pathology vector, it is possible to generate a series of images where the tumor size increases gradually, with the underlying brain structure remaining constant.

This experiment demonstrates that the model captures meaningful separation between anatomical and pathological features, as variations in the pathology vector result in controlled changes in tumor appearance while preserving structural consistency. By scaling up the pathology vector, it is possible to see features related to the tumor emerging gradually in the resulting MRI images, which is indicative of meaningful pathology information being learned by the generator, as opposed to arbitrary visual features. This clearly demonstrates that the model is allowing counterfactual reasoning to be performed on the generated tumors.

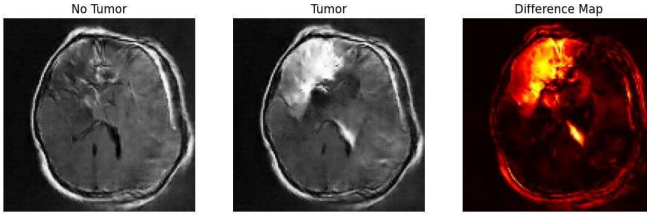


Fig. 7: Counterfactual Generation

The capacity to continuously modify the tumor intensity also allows for interactive visualization, which could potentially be used to illustrate the effect of the disease progression on the predictions made by the classifier and to further increase the overall level of transparency in medical AI systems.

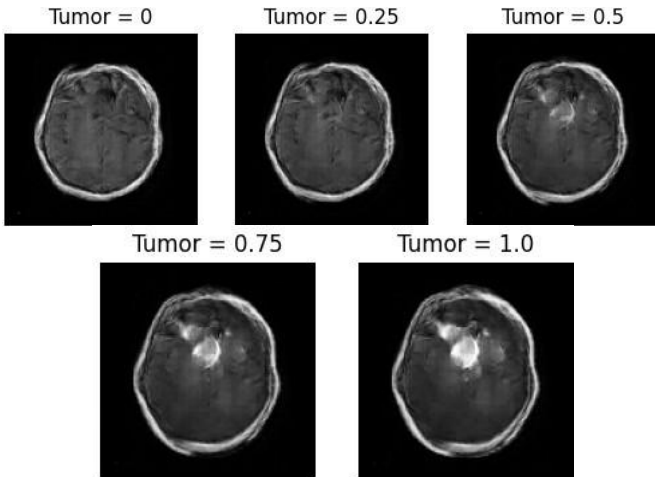


Fig. 8: Controllable Tumor Intensity Generation

The visualizations obtained using the Grad-CAM method also indicate that the classifier focuses on tumor regions, as the heat maps obtained using this method highlight the regions corresponding to the tumor location.

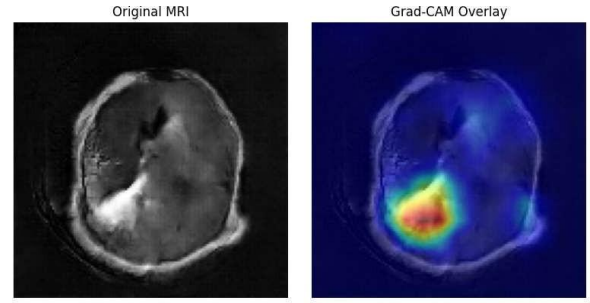


Fig. 9: GRAD-CAM Visual

To further verify the reliability of the explanation method, a masking-based evaluation experiment was also conducted, in which the regions highlighted in the image by the counterfactual difference map were removed, and the image was re-classified using the classifier [23]. If the regions highlighted in the image correspond to the tumor features, the overall confidence of the classifier in the presence of a tumor would significantly reduce.

In quantitative terms, the confidence of the classifiers decreased by about 41% using the masking approach, thereby validating the regions highlighted in the counterfactual difference maps as semantically relevant tumor characteristics.

V. CONCLUSION

In this paper, an explainable generation architecture has been introduced where the idea of latent disentanglement and classifier supervision has been employed for generating MRI of brain tumors in an interpretable way.

With the help of Grad-CAM explanation technique and counterfactual analysis, interesting observations have been made regarding tumor localization and the effect of features on it. The findings of this research prove that explainable methods can be effectively used with generative models in medical image analysis applications.

Such a suggested framework could be used to validate the usefulness of disentangled representation learning and classifier-guided generative modeling to enhance the explainability of the model. Furthermore, a new insight could be gained concerning the attributes related to brain tumors based on MRI images by making use of counterfactual visualization and pathological manipulation in this work. It is evident from the study's findings that the proposed framework is highly accurate in detecting brain tumors. While high accuracy has been achieved for the identification of brain cancers in this research, it should be mentioned that the measurement of brain tumor detection accuracy, precision, recall, and F1-score is the main emphasis of this work.

While additional assessment metrics like Learned Perceptual Image Patch Similarity (LPIPS), Fréchet Inception Distance (FID), and Structural Similarity Index Measure (SSIM) are used in other image synthesis tasks, FID score measures the distributional similarity, SSIM score measures the structural similarity, and LPIPS score measures the perceptual similarity. However, these metrics were not used in the proposed work because the focus of the work is on interpretation and control,

not on photorealism. Moreover, metrics such as FID score need a large dataset with a pre-trained feature extractor, which is not appropriate in the case of medical images since the feature extractor is trained on images that do not have domain-specific features. LPIPS score also uses perceptual features that are learned from images that do not have clinical relevance.

Therefore, metrics that were used in the proposed work were metrics that were used for classification-based evaluation. However, in future work, metrics that were used in GANs, namely FID, SSIM, and LPIPS, will be used in order to evaluate the quality of the generators. The proposed work is beneficial in building more transparent AI systems.

VI. FUTURE ENHANCEMENTS

The focus for future work will be on improving the realism and diversity of the generated MRI images by using more sophisticated GAN architecture [6] and medical imaging datasets. Expanding the model to support multi-class classification for tumor types and grades will improve its utility. Using 3D MRI volume data instead of 2D slices will provide richer anatomical context for tumor detection. Future research will investigate the application of attention mechanisms and transformers, which will improve classification performance and interpretation.

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